

# User Manual: IPA Data Processing Tool

*Convert Cartesian Coordinates from Individual Positions to a Unified Data File*

## 1 Introduction

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The **IPA Data Processing Tool** is a web-based application that prepares spatial data for terrestrial laser scanner (TLS) workflows. Specifically, it formats data collected during the OSAC Interim Performance Assessment (IPA) test procedure so it can be uploaded directly into the secondary **TLS Analysis Tool**.

During an IPA evaluation, four separate scans are taken of a stationary group of targets to collect their Cartesian  $(X, Y, Z)$  center coordinates. This process creates two main data management problems:

1. **Coordinate System Disconnect:** Each of the four scanning positions records data using its own local reference point. These individual coordinate systems must be kept separate.
2. **Scrambled Target Order:** The rows of data are often saved in a different, inconsistent order from scan to scan. However, the downstream **TLS Analysis Tool** requires all coordinates to be combined into a single file where the target rows match perfectly.

The **IPA Data Processing Tool** automatically solves these issues. The application matches the identical targets across all four scans, fixes the scrambled row orders, and combines the data into the exact single-file layout required by the **TLS Analysis Tool**.

## 2 Getting Started

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### 2.1 File Requirements & Constraints

All uploaded datasets must conform to the following standardized format:

- **File Format:** Plain text (ASCII encoded `.txt`) or comma-separated value (`.csv`) files only.
- **Required Files:** Exactly four (4) datasets must be supplied simultaneously, corresponding to the four scanning positions.
- **Dimensionality:** Each file must contain exactly three (3) columns representing the Cartesian coordinates  $(X, Y, Z)$ . No extraneous columns or qualitative tags are permitted within the data file.
- **Consistent Number of Targets:** All four uploaded files must have the exact same number of rows. This ensures that every target in the array was successfully measured from all four scanning positions.

- **Unique Filename:** Each file must have a unique name. Duplicate filenames will trigger an error.

### 3 User Interface Overview

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The application layout is divided into two primary sections:

1. **Control Sidebar (Left):** This section contains the file upload slots, data settings configuration, and the data processing execution controls.
2. **Main Display Panel (Right):** This section displays data validation messages, a preview of the processed output, and the download controls for the finished dataset.

### 4 Step-by-Step Operating Instructions

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#### 4.1 Step 1: Upload Files

Locate the left-hand configuration sidebar. Upload the coordinate files for Positions 1, 2, 3, and 4 into their matching upload slots. The target file order of Position 1 will serve as the master coordinate baseline to which all subsequent position indexes are dynamically remapped.

**Note:** The primary **Process Data** execution button will remain greyed out and unclickable until all four files have been uploaded and passed the unique naming check.

#### 4.2 Step 2: Configure Metadata and Naming Prefixes

Configure the data structure options located directly beneath the file upload inputs:

- **Header Configuration Toggle:** Set this dropdown to “Yes” if the data files contain a structural top header row (e.g., column label tags). If the numeric data starts directly on line 1, select “No”.
- **Target Naming Scheme Prefix:** Input a customized text string to define how target labels are compiled. The application automatically appends a sequential numeric suffix to this prefix until every target in the dataset has been uniquely identified. For example, entering a prefix of **Target\_** dynamically names the targets “**Target\_01**”, “**Target\_02**”, and so forth. Similarly, an input string of **CB0** automatically yields the sequential IDs “**CB001**” and “**CB002**”.

#### 4.3 Step 3: Process Data

Once the application checks and validates the files, the blue **Process Data** button becomes active. Click this button to run the application. The tool will automatically match the targets across all setups and combine the four separate files into a single, unified data file.

#### 4.4 Step 4: Error Handling & Validation

If the application identifies any issues with the data, processing stops immediately and a red error message appears at the top of the screen. Common errors include:

- **Header Mismatch:** This occurs if “No” is selected for headers, but the file actually contains a row of text or column names at the very top.

- **Missing Data (NA):** This occurs if any of the files contain blank spaces, missing numbers, or incomplete rows.
- **File Dimension Mismatch:** This occurs if the files do not contain exactly three columns ( $X, Y, Z$ ) or if the files do not all contain the exact same number of rows.

**TROUBLESHOOTING NOTE:** If data validation fails, check the original data files for empty spaces, character padding, or truncated rows before attempting to process the data again.

## 4.5 Step 5: Review Output and Export Data

If the data passes all checks, a summary appears showing the total number of matched targets. A preview table also displays the consolidated data, organized to match the exact target sequence of Position 1.

After verifying that the table contains the expected number of rows and that the **TargetID** column displays the desired naming prefix, click the green **Download Processed Data** button at the bottom of the screen. The consolidated data is saved to the computer as a standard comma-separated values file named `processed_data_YYYY-MM-DD.csv`.

## 4.6 Resetting the Application

To clear the environment and start over with a fresh set of data, click the red **Reset App** button at the bottom of the sidebar to refresh the application instantly.

# 5 Example Datasets

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Two sets of example files are provided with the **IPA Data Processing Tool** to show how the application handles different file types. Each set contains four separate position files:

- **CSV Example Set:** Comma-separated value (`.csv`) files that include column headers at the top. When using these files, the **Header Configuration Toggle** must be set to “Yes”.
- **TXT Example Set:** Plain text (`.txt`) files that do not include headers. When using these files, the **Header Configuration Toggle** must be set to “No”.

Processing the four files from either example set through the **IPA Data Processing Tool** creates a single, combined data file. This output file matches the exact structure of the `LSS999_Jan2026.csv` file found within the `ExampleData2/January2026` folder of the secondary **Data Analysis Tool**.